

Git, Part 2: The Humans

User Study Day · Do Developers Really Know How to Use Git Commands? A Large-Scale Study Using Stack Overflow

The Humans

Open the class link (it ends in /thu) and type your first name.

Do Developers Really Know How to Use Git Commands? A Large-Scale Study Using Stack Overflow

Yang, Zhang, Pan, Xu, Zhou, Huang · ACM TOSEM 2022

Take the paper's survey

The real questions, from the authors' published form.

92 developers answered them. You are next.

2. What is your professional area?
3. What is your highest degree?
4. How many years have you been using Git?
5. Which of the following levels do you consider your ability to use Git to be?
6. What are the main approaches you used to learn Git commands?
7. Do you have any suggestions for potential new developers on learning Git commands

Q1 of the form, the donation choice, is left out.

1 · THE QUESTION

The paper in one slide

RQ1 · How many Git questions?	80,370 questions, 2008–2020. Since 2010, about 0.4% of all Stack Overflow questions each year.
RQ2 · Who asks?	In 2020, 40.0% of Git askers had registered more than 5 years earlier (21.2% of all askers).
RQ3 · Which commands?	Most viewed on average, among commands in 200+ questions: <code>git revert</code> , <code>git reflog</code> , <code>git stash</code> , <code>git clean</code> , <code>git reset</code> .
RQ4 · Which are hardest?	Most often without an accepted answer: rare commands; among frequent ones, <code>git credential</code> and <code>git submodule</code> .
RQ5 · How do people learn?	Survey, 92 of 508 invited: 81.7% of ticked approaches are self-learning.

What kind of study is this?

Need-finding: what problems do people have?

Formative: which solution looks promising?

Evaluative: did our tool work?

1 · THE QUESTION

Need-finding

Need-finding: what problems do people have? ✓

Formative: which solution looks promising?

Evaluative: did our tool work?

Its shape: what problems do developers face when using Git commands?

A need-finding study must see what goes badly **and** what goes well. Stack Overflow questions record what people got stuck on; when Git just worked, nobody asked.

Is the question interesting?

Stack Overflow posts are...

Observation: someone watched people use Git

Traces: records people left while working

Self-report: people describing their own experience

Traces. Of asking.

Observation: someone watched people use Git

Traces: records people left while working ✓

Self-report: people describing their own experience

Sarah's ladder: watching beats traces, traces beat asking. Stack Overflow questions are on her list of traces.

Traces of asking, not of using. A question body is also partly self-report: the asker describes the problem.

RQ5 is a survey. Nobody in this paper was watched using Git.

Can this data answer “do developers know how to use Git commands?”

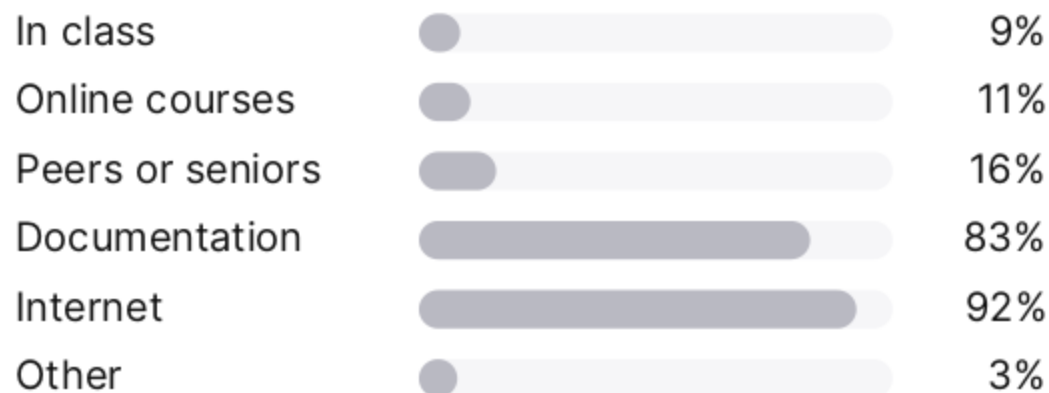
2 · THE DATA

You and the paper's 92

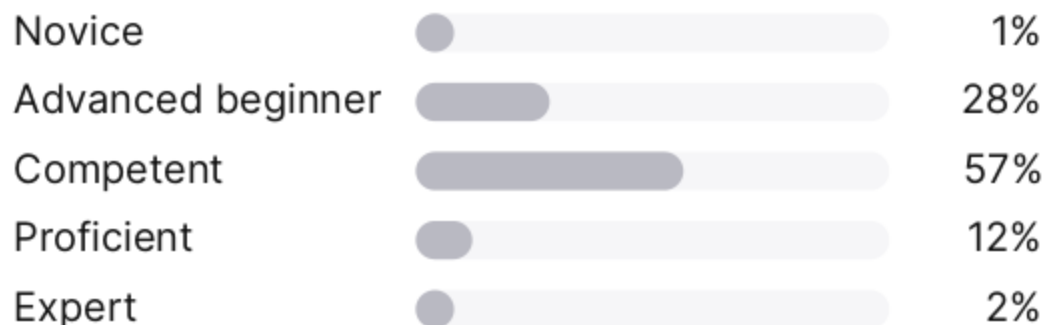
81.7% is 161 of 197 ticked boxes. By people: 85 of 92 ticked the internet.

Invited: developers who had recently asked a Git question on Stack Overflow, plus some researchers. 92 of 508 answered.

THE PAPER'S 92: HOW THEY LEARNED (SHARE OF PEOPLE)



THE PAPER'S 92: GIT LEVEL, SELF-RATED



Median years using Git: 8. In class, the app shows your answers next to these.

Your level, and how you learned: which would you trust?

Check the rule on 8 real posts

The paper's rule: a post counts for every Git command in its question **or its accepted answer**.

For each post: is the asker stuck on that command, do they need it without knowing its name, or is it not about it?

STUCK ON IT

They ask about this command.

NEEDS IT, CAN'T NAME IT

It would solve their problem; they don't know it.

NOT ABOUT IT

It is only mentioned.

On your laptop: 8 posts, each with the question, the accepted answer, and the command the paper counted it for.

Drawn at random from short, answered posts credited to the top-5 commands (analysis/thursday_numbers.py, seed 294).

3 · THE ANALYSIS

The rule and you

The rule counts all 8 posts the same. In 5, only the accepted answer names the command.

“Stuck on it” is evidence a command is confusing. “Needs it, can’t name it” supports the paper’s own idea of recommending commands (§4.1).

POST	COUNTED FOR	ASKER NAMES IT
git how to move back commit (6,163 views)	git reflog	Yes
How do I remove a single file from the staging area (undo git add)? (656,913 views)	git reset	Answer only
Undo a Git merge that hasn't been pushed yet (2,371,183 views)	git reflog	Answer only
How can i undo changes between two commits? (24 views)	git revert	Answer only
git 'merge' only existing files (1,378 views)	git clean	Yes
Git branch not working the way I expected it to (2,989 views)	git reset	Yes
revert a commit after push, checkout a commit behind and modification of files (46 views)	git revert	Answer only
Revert erroneous git checkout (45 views)	git stash	Answer only

Did they measure “developers find this command hard”?

Code the comments yourself

The paper's Table 8: seven survey comments, each placed in one of six categories.

Put each comment where you would. Then we compare with the paper, and with each other.

The paper's six categories:

Learning and improving from practice (18)

Making the best use of the internet (17)

Understanding basic concepts and principles first (14)

Emphasizing the importance of Git (9)

Mastering only the basic commands (5)

Having the basic needs satisfied by Git GUI tools (2)

3 · THE ANALYSIS

Coding without a method

The paper sorted 65 comments into 6 categories “for better delivery purposes”. It names no method and reports no second coder or agreement.

It did measure agreement elsewhere: Cohen’s kappa 0.844 for which posts are about Git commands (§2.1).

COMMENT	THE PAPER’S CATEGORY
#56 “Git is an extremely important tool to our routine as software ...”	Emphasizing the importance of Git
#32 “There are so many commands, I think it is better to learn them in ...”	Learning and improving from practice
#89 “Git is not a tool you have to go pretty deep to learn. Since it is a ...”	Mastering only the basic commands
#26 “The difficulty with git is that when problems arise, you have to dive ...”	Understanding basic concepts and principles first
#21 “All I want is to concentrate on my development, and dump the code ...”	Having the basic needs satisfied by Git GUI tools
#77 “Git is my VCS of choice, as I’ve learned how it works and have use ...”	Learning and improving from practice
#42 “Years on, I still constantly have to search the internet for help ...”	Making the best use of the internet

What would make Table 8 trustworthy?

BREAK

Break

5 minutes. Then groups.

Put one claim on trial

Your group gets one of the paper's claims. One person types.

Write what they measured, then the strongest sentence the data supports.

RQ2 ANSWER, §3.2

"This suggests that even developers with years of development experience can have trouble using Git commands."

§1, FINDING (4); RQ4, §3.4

"...for the more frequently-used commands, git credential and git submodule are among the most difficult ones."

ABSTRACT; RQ5, §3.5

"Self-learning is the primary learning approach."

§3.5

"This result, although surprising, is consistent with the conclusion we obtained in RQ2, indicating that even experienced developers still have doubts about Git usage."

4 · DO YOU BELIEVE IT?

The verdicts (1/2)

The title asks whether developers know how to use Git commands. Nothing in the paper watches anyone use one.

"This suggests that even developers with years of development experience can have trouble using Git commands."

THE DATA SUPPORTS

As written, it holds: some long-registered askers, including the two quoted, had trouble.

"...for the more frequently-used commands, git credential and git submodule are among the most difficult ones."

THE DATA SUPPORTS

git credential's questions lack an accepted answer more often than Git questions overall (paper: 43.0%; our re-run: 39.3% vs 36.5%). git submodule's barely differ (37.5%).

Which claim survives best, and why?

4 · DO YOU BELIEVE IT?

The verdicts (2/2)

The title asks whether developers know how to use Git commands. Nothing in the paper watches anyone use one.

"Self-learning is the primary learning approach."

THE DATA SUPPORTS

Most of these 92 say they learned from the internet (85) and the documentation (76).

"This result, although surprising, is consistent with the conclusion we obtained in RQ2, indicating that even experienced developers still have doubts about Git usage."

THE DATA SUPPORTS

Most of these 92, many with more than five years of Git, rate themselves competent or below (79 of 92).

Which claim survives best, and why?

Design for the need that survived

The need: undo. The most-viewed Git question asks how to undo the most recent local commits. Its accepted answer's fix is `git reset HEAD~`; its further reading points to `git reflog`.

Why do people have to ask? Design one change, and the study that would show it works.

ONE CHANGE TO GIT

What the user sees or types

THE STUDY

Does ...
increase/decrease ... for
... doing ...?

MEASURE, AND ONE THREAT

e.g. time to recover;
threat: a learning effect

YOUR DATA

Watch people, collect
traces, or ask them?
Which, exactly?

Your designs

Each group reads its change, its study, its measure and threat, and its data. The app shows all of them.

Which of these could you test by watching people? Which threat is hardest?

What Git and Jujutsu did

Git 2.23 (2019) split `git checkout` into `git switch` and `git restore`. In 2020 a developer “working with Git for years” asked what `git checkout [file]` does next to `git restore`: the paper’s reference [41].

Jujutsu (`jj`) logs every operation, and `jj undo` reverses the last one.

Which data would convince Git’s maintainers your change works?

The paper: what holds, what doesn't

Holds

- ✓ Git questions are many, and steady: about 0.4% of Stack Overflow a year.
- ✓ Undo is a real need: the most-viewed Git question asks how to undo a commit.
- ✓ Experienced developers can have trouble: true, as written.
- ✓ The data is public, so anyone can check it. We did.

Does not hold, as written

- × “Hardest commands”: the rule counts fixes in answers as problems.
- × The command ranking: one post moves git reflog from #8 to #2.
- × “Self-learning is primary”: asked of people found on Stack Overflow.
- × Table 8 and “doubts”: no coding method; self-ratings, not skill.
- × The title’s question: nobody was watched using Git.

EXIT

2 min

One new idea for building tools for people

One line, on your own.

THURSDAY

Asking is not using.

Neither the survey (we ask them) nor Stack Overflow (they ask) shows anyone using Git.

Both days: check what your data actually records before you believe it.